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END-TO-END PIZZA SALES ANALYSIS USING SQL

WHERE EVERY SLICE TELLS A STORY

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Performed comprehensive end-to-end data analysis on pizza sales data using advanced SQL techniques including joins, aggregations, CTEs, and window functions to identify revenue trends and top-performing products.

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ESSENTIAL INSIGHTS TO EVALUATE PIZZA SALES

Basic:

- *Retrieve the total number of orders placed.*
- *Calculate the total revenue generated from pizza sales.*
- *Identify the highest-priced pizza.*
- *Identify the most common pizza size ordered.*
- *List the top 5 most ordered pizza types along with their quantities.*

Intermediate:

- *Join the necessary tables to find the total quantity of each pizza category ordered.*
- *Determine the distribution of orders by hour of the day.*
- *Join relevant tables to find the category-wise distribution of pizzas.*
- *Group the orders by date and calculate the average number of pizzas ordered per day.*
- *Determine the top 3 most ordered pizza types based on revenue.*

Advanced:

- *Calculate the percentage contribution of each pizza type to total revenue.*
- *Analyze the cumulative revenue generated over time.*
- *Determine the top 3 most ordered pizza types based on revenue for each pizza category.*

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RETRIEVE THE TOTAL NUMBER OF ORDERS PLACED.

```sql
SELECT
    COUNT(*) AS no_of_orders_placed
FROM
    order_details;
```

	no_of_orders_placed
▶	48620

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CALCULATE THE TOTAL REVENUE GENERATED FROM PIZZA SALES.

```
SELECT
    ROUND(SUM(p.price * o.quantity), 0) AS total_revenue
FROM
    pizzas p
        JOIN
    order_details o ON p.pizza_id = o.pizza_id;
```

total_revenue
817860

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IDENTIFY THE HIGHEST-PRICED PIZZA.

```
SELECT
    pt.name, p.price AS highest_priced_pizza
FROM
    pizzas p
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
WHERE
    p.price = (SELECT
            MAX(price)
        FROM
            pizzas);
```

```
SELECT
    pt.name, p.price AS highest_priced_pizza
FROM
    pizzas p
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;
```

name	highest_priced_pizza
The Greek Pizza	35.95

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IDENTIFY THE MOST COMMON PIZZA SIZE ORDERED.

```sql
SELECT
    p.size, COUNT(o.order_details_id) AS order_quantity
FROM
    order_details o
        JOIN
    pizzas p ON o.pizza_id = p.pizza_id
GROUP BY p.size
ORDER BY order_quantity DESC
LIMIT 1;
```

size	order_quantity
L	18526

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LIST THE TOP 5 MOST ORDERED PIZZA TYPES ALONG WITH THEIR QUANTITIES.

```
SELECT
    p.pizza_type_id, pt.name, sum(o.quantity) AS total_quantity
FROM
    order_details o
        JOIN
    pizzas p ON o.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON pt.pizza_type_id = p.pizza_type_id
GROUP BY p.pizza_type_id , pt.name
ORDER BY total_quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Ce

pizza_type_id	name	total_quantity
classic_dlx	The Classic Deluxe Pizza	2453
bbq_ckn	The Barbecue Chicken Pizza	2432
hawaiian	The Hawaiian Pizza	2422
pepperoni	The Pepperoni Pizza	2418
thai_ckn	The Thai Chicken Pizza	2371

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JOIN THE NECESSARY TABLES TO FIND THE TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED.

```sql
SELECT
    pt.category, sum(o.quantity) AS total_quantity
FROM
    order_details o
        JOIN
    pizzas p ON o.pizza_id = p.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.category;
```

Result Grid | Filter Rows:

category	total_quantity
Classic	14888
Veggie	11649
Supreme	11987
Chicken	11050

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DETERMINE THE DISTRIBUTION OF ORDERS BY HOUR OF THE DAY.

```sql
SELECT
    HOUR(order_time) as hour, COUNT(*) AS total_orders
FROM
    orders o
GROUP BY HOUR(order_time)
ORDER BY total_orders DESC;
```

Result Grid | Filter Row

hour	total_orders
12	2520
13	2455
18	2399
17	2336
19	2009
16	1920
20	1642
14	[illegible]
15	1468
11	1231
21	1198
22	663

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JOIN RELEVANT TABLES TO FIND THE CATEGORY-WISE DISTRIBUTION OF PIZZAS.

```sql
SELECT
    pt.category AS pizza_category,
    COUNT(*) AS total_no_of_pizzas
FROM
    pizza_types pt
GROUP BY pt.category;
```

Result Grid | Filter Rows:

pizza_category	total_no_of_pizzas
Chicken	6
Classic	8
Supreme	9
Veggie	9

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GROUP THE ORDERS BY DATE AND CALCULATE THE AVERAGE NUMBER OF PIZZAS ORDERED PER DAY.

```sql
SELECT
    ROUND(AVG(daily_total), 0) AS avg_pizzas_per_day
FROM
    (SELECT
        o.order_date AS day, SUM(od.quantity) AS daily_total
    FROM
        orders o
    JOIN order_details od ON o.order_id = od.order_id
    GROUP BY o.order_date) AS daily_orders;
```

avg_pizzas_per_day
138

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DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE.

```sql
SELECT
    pt.name, SUM(p.price * od.quantity) AS revenue
FROM
    pizzas p
        JOIN
    order_details od ON p.pizza_id = od.pizza_id
        JOIN
    pizza_types pt ON p.pizza_type_id = pt.pizza_type_id
GROUP BY pt.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Exp

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

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CALCULATE THE PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE.

```sql
with total_revenue as (SELECT
    SUM(p.price * o.quantity) AS total_sales
FROM
    pizzas p
        JOIN
    order_details o ON p.pizza_id = o.pizza_id)
select pt.category,
round(sum(p.price * od.quantity)/(SELECT total_sales FROM total_revenue) * 100, 0)
as percentage_contribution
from pizza_types pt
join pizzas p on pt.pizza_type_id = p.pizza_type_id
join order_details od on od.pizza_id = p.pizza_id
group by pt.category;
```

```sql
-- or by using window function

select pt.category,
round(
    sum(p.price * od.quantity)
    /sum(sum(p.price * od.quantity)) over()
    * 100, 0
    ) as percentage_contribution
from pizza_types pt
join pizzas p on pt.pizza_type_id = p.pizza_type_id
join order_details od on od.pizza_id = p.pizza_id
group by pt.category;
```

Result Grid | Filter Rows:

category	percentage_contribution
Classic	27
Veggie	24
Supreme	25
Chicken	24

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ANALYZE THE CUMULATIVE REVENUE GENERATED OVER TIME.

```sql
select date, daily_revenue, sum(daily_revenue) over(order by date) as cum_revenue
from
(SELECT
    o.order_date as date, round(sum(p.price * od.quantity),0) as daily_revenue
FROM
    orders o
    join order_details od on o.order_id = od.order_id
    join pizzas p on p.pizza_id = od.pizza_id
GROUP BY o.order_date
ORDER BY o.order_date) as sales;
```

Result Grid Filter Rows:

date	daily_revenue	cum_revenue
2015-01-01	2714	2714
2015-01-02	2732	5446
2015-01-03	2662	8108
2015-01-04	1755	9863
2015-01-05	2066	11929
2015-01-06	2429	14358
2015-01-07	2202	16560
2015-01-08	2838	19398
2015-01-09	2127	21525
2015-01-10	2464	23989
2015-01-11	1872	25861
2015-01-12	1919	27780

Result 5

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DETERMINE THE TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE FOR EACH PIZZA CATEGORY.

```
select *
from
(select category, name, revenue,
dense_rank() over(partition by category order by revenue desc) as pizza_rank
from
(select pt.category, pt.name, round(sum(p.price * od.quantity),0) as revenue
from
    pizzas p
    join order_details od on p.pizza_id = od.pizza_id
    join pizza_types pt on pt.pizza_type_id = p.pizza_type_id
group by pt.name,  pt.category) sales ) ranked
where pizza_rank < 4;
```

Result Grid | Filter Rows: | Export: | Wrap

category	name	revenue	pizza_rank
Chicken	The Thai Chicken Pizza	43434	1
Chicken	The Barbecue Chicken Pizza	42768	2
Chicken	The California Chicken Pizza	41410	3
Classic	The Classic Deluxe Pizza	38180	1
Classic	The Hawaiian Pizza	32273	2
Classic	The Pepperoni Pizza	30162	3
Supreme	The Spicy Italian Pizza	34831	1
Supreme	The Italian Supreme Pizza	33477	34831
Supreme	The Sicilian Pizza	30940	3
Veggie	The Four Cheese Pizza	32266	1
Veggie	The Mexicana Pizza	26781	2
Veggie	The Five Cheese Pizza	26066	3

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Thank you for reviewing this analysis. This project reflects my ability to apply SQL to real-world business problems, and I look forward to leveraging these skills in professional data roles.